

Mathematical Induction

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TROCKERS

SYLLABUS (6042) REQUIREMENTS

- describe the process of mathematical induction
- prove by mathematical induction to establish a given result
- use the strategy of conducting limited trials to formulate a conjecture and proving it by the method of induction

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MATHEMATICAL INDUCTION

Definition: It is a method of proof in which a statement is proved for one step in a process, and it is shown that if the statement holds for that step, it holds for the next.

STEPS

i. **Base Case/Initial Stage**

Show it is true for $n = 1, n = 2$ etc and give a conclusion for this stage.

ii. **Assumption/Hypothesis Stage**

Suppose/ assume the statement is true for $n = k$.

iii. **Thesis Stage**

Stating the result for $n = k + 1$.

iv. **Proof Stage**

Proving true for $n = k + 1$. Use the thesis stage and the hypothesis stage to get the required result. Give a conclusion for this stage.

v. **Conclusion Stage**

State four conditions clearly: $n = 1, n = k, n = k + 1$ and $n \in \mathbb{Z}^+$

Areas to consider

- Divisibility
- Series/Summations
- Matrices
- Integration
- Differentiation
- Factorial Notation
- Composite Functions
- Binomial etc

DIVISIBILITY

PROOF HINT

For example to prove that $x^n + y$ is divisible by z :

Method 1:

(i) If the statement is denoted by P_n then:

Use $P_{k+1} - P_k$ (Since subtracting consecutive multiples of a certain number gives the same number e.g. $18 - 15 = 3$ or $33 - 30 = 3$)

OR

Method 2:

(ii) Assumption $x^k + y = zA$

$$\Rightarrow x^k = zA - y$$

Proof $x^{k+1} + y = zB$

Apply the laws of indices and substitute the Assumption i.e.

$$\begin{aligned}x^{k+1} + y &= x^k \times x + y \\&= x(x^k) + y \\&= x(zA - y) + y\end{aligned}$$

Examples

Question 1

Prove that $4^n + 5$ is divisible by 3 for all $n \in \mathbb{Z}^+$.

Suggested Solution

When $n = 1$:

$4^1 + 5 = 9$, which is divisible by 3 or we can say 9 is a multiple of 3.

$\therefore$ The statement is true for $n = 1$. (Base Case/Initial Stage)

Assuming true for $n = k$:

$$4^k + 5 = 3A \Rightarrow 4^k = 3A - 5. \text{ (Assumption/Hypothesis Stage)}$$

Then for $n = k + 1$: $4^{k+1} + 5 = 3B$. (Thesis Stage)

Proof

$$4^{k+1} + 5 = 3B$$

LHS:

$$\begin{aligned} 4^{k+1} + 5 &= 4^k \cdot 4^1 + 5 \\ &= 4^1 \cdot 4^k + 5 \\ &= 4(3A - 5) + 5 \\ &= 4 \cdot 3A - 20 + 5 \\ &= 4 \cdot 3A - 15 \\ &= 3(4A - 5) \\ &= 3B \text{ where } B = 4A - 5; \text{ (which is true for } n = k + 1) \end{aligned}$$

$\therefore$ The statement is true for $n = k + 1$. (Proof Stage)

NB: Alternative Proof Method (Not applicable to all problems)

$P_{k+1} - P_k$ is also a multiple of 3

$$4^{k+1} + 5 - (4^k + 5)$$

$$4^k(4^1) + 5 - 4^k - 5$$

$$4^k(4^1) + 5 - 4^k - 5$$

$$4(4^k) - 4^k$$

$3(4^k)$, which is divisible by 3

$\therefore$ The statement is true for $n = k + 1$. (Proof Stage)

Since the statement is true for $n = 1, n = k, n = k + 1 \therefore$ by mathematical induction the statement is also true for $n \in \mathbb{Z}^+$ (Conclusion Stage)

HINT

$6^n - 1$ is divisible by 5

$5^n - 1$ is divisible by 4

$4^n - 1$ is divisible by 3

$\vdots$

$k^n - 1$ is divisible by $(k - 1)$

Also

$6^n + 1$ is divisible by 7

$5^n + 1$ is divisible by 6

$4^n + 1$ is divisible by 5

$\vdots$

$k^n + 1$ is divisible by $(k + 1)$

Question 2

ZIMSEC JUNE 2013 PAPER 2

Prove by induction that $9^{2n} - 5^{2n}$ is divisible by 7 for $n \in \mathbb{Z}^+$

Suggested Solution

When $n = 1$:

$$\begin{aligned}9^{2n} - 5^{2n} &\equiv 9^2 - 5^2 \\&= 81 - 25 \\&= 56 \\&= 7 \times 8 \text{ which is divisible by } 7\end{aligned}$$

$\therefore$ The statement holds for $n = 1$.

Assuming true for $n = k$:

$$9^{2k} - 5^{2k} = 7A \Rightarrow 9^{2k} = 7A + 5^{2k}.$$

Then for $n = k + 1$: $9^{2(k+1)} - 5^{2(k+1)} = 7B$

Proof

$$9^{2(k+1)} - 5^{2(k+1)} = 7B$$

LHS:

$$\begin{aligned}9^{2(k+1)} - 5^{2(k+1)} &= 9^{2k} \times 9^2 - 5^{2k} \times 5^2 \\&= 81(9^{2k}) - 25(5^{2k}) \\&= 81(7A + 5^{2k}) - 25(5^{2k}) \\&= 81(7A) + 81(5^{2k}) - 25(5^{2k}) \\&= 81(7A) + 56(5^{2k}) \\&= 7[81A + 8(5^{2k})] \\&= 7B \text{ where } B = 81A + 8(5^{2k}) \text{ (which is true for } n = k + 1\text{)}\end{aligned}$$

$\therefore$ The statement holds for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1 \therefore$ by mathematical induction the statement holds for $n \in \mathbb{Z}^+$

Question 3

Prove using the principles of mathematical induction that $7^n - 2^n$ is divisible by 5 for all $n \in \mathbb{N}$

Suggested Solution

When $n = 1$:

$$7^n - 2^n = 7^1 - 2^1 = 7 - 2 = 5 \text{ (which is a multiple of 5)}$$

$\therefore$ The statement is true for $n = 1$.

Assuming true for $n = k$:

$$\begin{aligned} 7^k - 2^k &= 5A \\ \Rightarrow 7^k &= 5A + 2^k \end{aligned}$$

Then for $n = k + 1$:

$$7^{k+1} - 2^{k+1} = 5B$$

Proof

$$7^{k+1} - 2^{k+1} = 5B$$

LHS

$$\begin{aligned} 7^{k+1} - 2^{k+1} &= 7(7^k) - 2(2^k) \\ &= 7(5A + 2^k) - 2(2^k) \\ &= 7(5A) + 7(2^k) - 2(2^k) \\ &= 7(5A) + 5(2^k) \\ &= 5(7A + 2^k) \approx 5B \text{ (which is true for } n = k + 1) \end{aligned}$$

$\therefore$ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for all $n \in \mathbb{N}$.

Question 4

Prove by induction that $3^n + 7^n$ is divisible by 10 for all odd $n \in \mathbb{Z}^+$

Suggested Solution

When $n = 1$:

$$\begin{aligned}3^n + 7^n &\equiv 3^1 + 7^1 \\&= 10 \\&= 10 \times 1 \text{ which is divisible by } 10\end{aligned}$$

$\therefore$ The statement holds for $n = 1$.

Assuming true for $n = k$:

$$3^k + 7^k = 10A \Rightarrow 3^k = 10A - 7^k.$$

Remember the sequence for odd numbers 1, 3, 5 ... i.e. $n, n+2, n+4, \dots$

$$\text{Then for } n = k+2: 3^{k+2} + 7^{k+2} = 10B$$

Proof

$$3^{k+2} + 7^{k+2} = 10B$$

LHS:

$$\begin{aligned}3^{k+2} + 7^{k+2} &= 3^k \times 3^2 + 7^k \times 7^2 \\&= 9(3^k) + 49(7^k) \\&= 9(3^k) + 49(7^k) \\&= 9(10A - 7^k) + 49(7^k) \\&= 9(10A) - 9(7^k) + 49(7^k) \\&= 9(10A) + 40(7^k) \\&= 10[9A + 4(7^k)] \\&= 10B \text{ where } B = 9A + 4(7^k) \text{ (which is true for } n = k+2\text{)}\end{aligned}$$

$\therefore$ The statement holds for $n = k+2$.

Since the statement is true for $n = 1, n = k, n = k+2 \therefore$ by mathematical induction the statement holds for all odd $n \in \mathbb{Z}^+$

PRACTICE QUESTIONS

Question 1

Prove that $5^n + 2(11^n)$ is a multiple of 3 for all positive integers of n .

Question 2

Prove that $49^n + 16n - 1$ is divisible by 64 for all natural numbers n .

Question 3

Prove that $6^n + 8^n$ is a multiple of 7, when n is odd.

Question 4

Prove by mathematical induction that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for $n \in \mathbb{N}$

Question 5

Prove using the principles of mathematical induction that $7^n - 2^n$ is divisible by 5 for $n \in \mathbb{N}$

Question 6

Prove that $2^{6n} + 3^{2n-2}$ is a divisible of 5 for all positive integers of n .

Question 7

Prove using the principle of mathematical induction that $7^{2n} - 3 \times 5^n + 2$ is divisible by 7 for $n \in \mathbb{N}$

Question 8

Prove by mathematical induction that $n(n + 2)$ is divisible by 4 for all even numbers: $n \geq 2$

Question 9

Prove using mathematical induction that $10^{2n-1} + 1$ is divisible by 11 for $n \in \mathbb{N}$

Question 10

Prove by mathematical induction that $2^{n+2} + 3^{2n+1}$ is divisible by 7 for $n \in \mathbb{N}$

Question 11

Prove by mathematical induction that $3^{2n+2} - 8n - 9$ is divisible by 8 for $n \in \mathbb{N}$

Question 12

Prove by mathematical induction that $3^{2n+1} + 2^{n+2}$ is divisible by 7 for $n \in \mathbb{Z}^+$

Question 13

Prove using mathematical induction that $3^{2n} + 11$ is divisible by 4 for $n \in \mathbb{N}$

Question 14

Prove by mathematical induction that $n^3 + 7n + 9$ is divisible by 3 for $n \in \mathbb{N}$

Question 15

Prove by mathematical induction that $5^{2n} + 7$ is a multiple of 8 for $n \geq 1$

Question 16

Prove by induction that $5^{2n} - 1$ is a multiple of 8 for $n \in \mathbb{N}$

Question 17

Prove by mathematical induction that $10^{3n} + 13^{n+1}$ is divisible 7 for $n \geq 1$

Question 18

Prove by mathematical induction that $2^{3n} - 3^n$ is divisible 5 for $n \geq 1$

PAST EXAMINATION QUESTIONS

Question 1

ZIMSEC NOVEMBER 2005 PAPER 2

Prove by induction that $3^n - 2n - 1$ is divisible by 4 for $n \in \mathbb{Z}^+$. [6]

Question 2

ZIMSEC NOVEMBER 2006 PAPER 2

Prove that the number $U_n = 6^n + 4$ is divisible by 5 for all positive integral values n . [6]

Question 3

ZIMSEC NOVEMBER 2009 PAPER 2

Prove by induction that $5^{2n} - 3^{2n}$ is divisible by 8 for $n \in \mathbb{Z}^+$. [7]

Question 4

ZIMSEC NOVEMBER 2010 PAPER 2

Prove by mathematical induction that $7^n - 6n - 1$ is divisible by 36 for $n \in \mathbb{N}, n > 1$. [6]

Question 5

ZIMSEC JUNE 2018 PAPER 2

Prove by induction that a number $U_n = 2^{6n} + 3^{2n-2}$ is divisible by 5 for all positive integral value of n . [7]

Question 6

ZIMSEC NOVEMBER 2018 PAPER 2

Prove by mathematical induction that for all positive integers, n , $10^n + 3 \times 4^{n+1} + 5$ is divisible by 3. [6]

SUMMATIONS/SERIES

PROOF HINT

Remember that

$$\sum_{r=1}^k r = 1 + 2 + 3 + \cdots + (k-2) + (k-1) + k$$

and that

$$\sum_{r=1}^{k+1} r = 1 + 2 + 3 + \cdots + (k-2) + (k-1) + k + (k+1)$$

$$= \sum_{r=1}^k r + (k+1)$$

$\therefore$ Add the assumption stage and the $(k+1)^{\text{th}}$ stage

Examples

Question 1

Prove by induction that:

$$\sum_{r=1}^n r = \frac{n}{2}(n+1) \text{ for all } n \in \mathbb{Z}^+$$

Suggested Solution

When $n = 1$:

$$LHS = 1$$

$$RHS = \frac{1}{2}(1+1) = \frac{1}{2}(2) = 1 \equiv LHS$$

Since $RHS \equiv LHS = 1 \therefore$ the statement is true for $n = 1$.

Assuming that the statement is true for $n = k$:

$$\sum_{r=1}^k r = \frac{k}{2}(k+1)$$

Then for $n = k + 1$:

$$\sum_{r=1}^{k+1} r = \frac{k+1}{2}[(k+1)+1]$$

Proof

Note:

$$\text{a) } \sum_{r=1}^k r = 1 + 2 + 3 + \dots + k$$

$$\text{b) } \sum_{r=1}^{k+1} r = 1 + 2 + 3 + \dots + k + (k+1) = \sum_{r=1}^k r + (k+1)$$

$$\sum_{r=1}^{k+1} r = \frac{k+1}{2}[(k+1)+1]$$

LHS:

$$\begin{aligned} \sum_{r=1}^{k+1} r &= \sum_{r=1}^k r + (k+1) \\ &= \frac{k}{2}(k+1) + (k+1) \\ &= (k+1)\left(\frac{k}{2} + 1\right) \\ &= (k+1)\left(\frac{k+2}{2}\right) \\ &= (k+1)\left[\frac{(k+1)+1}{2}\right] \\ &= \frac{k+1}{2}[(k+1)+1], \text{ (which is true for } n = k+1) \end{aligned}$$

∴ The statement is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1$ ∴ by mathematical induction the statement holds for all positive integral values of n .

Question 2

Prove by induction that:

$$1 + 4 + 7 + \dots + (3n - 2) = \frac{n(3n-1)}{2} \text{ for all positive integral values of } n.$$

Suggested Solution

When $n = 1$:

$$LHS = 3(1) - 2 = 3 - 2 = 1$$

$$RHS = \frac{1 \cdot [3(1) - 1]}{2} = \frac{1(2)}{2} = \frac{2}{2} = 1 \equiv LHS$$

Since $RHS \equiv LHS = 1$ ∴ P_n is true for $n = 1$.

Assuming true for $n = k$:

$$1 + 4 + 7 + \dots + (3k - 2) = \frac{k(3k - 1)}{2}$$

Then for $n = k + 1$:

$$1 + 4 + 7 + \dots + [3(k + 1) - 2] = \frac{(k + 1)[3(k + 1) - 1]}{2}$$

Proof

Note:

a) The summation of k terms $= 1 + 4 + 7 + \dots + (3k - 2)$

b) The summation of $(k+1)$ terms

$$= 1 + 4 + 7 + \dots + (3k - 2) + [3(k + 1) - 2]$$

$$\begin{aligned}
 &= 1 + 4 + 7 + \cdots + (3k - 2) + (3k + 1) \text{ (Bracket Expansion)} \\
 &= \text{the summation of } k \text{ terms } [1 + 4 + 7 + \cdots + (3k - 2)] + \\
 &\quad \text{the } (k + 1)^{\text{th}} \text{ term } [3k + 1]
 \end{aligned}$$

$$\begin{aligned}
 1 + 4 + 7 + \cdots + [3(k + 1) - 2] &= 1 + 4 + 7 + \cdots + (3k + 1) \\
 &= 1 + 4 + 7 + \cdots + (3k - 2) + (3k + 1) \\
 &= \frac{k}{2}(3k - 1) + (3k + 1)
 \end{aligned}$$

$$= \frac{k(3k - 1) + 2(3k + 1)}{2}$$

$$= \frac{3k^2 - k + 6k + 2}{2}$$

$$= \frac{3k^2 + 5k + 2}{2}$$

$$= \frac{(k + 1)(3k + 2)}{2} \text{ (Factorisation)}$$

$$= \frac{(k + 1)[(3k + 3) - 1]}{2}$$

$$= \frac{(k + 1)[3(k + 1) - 1]}{2} \text{ (which is true for } n = k + 1 \text{)}$$

$\therefore$ The statement is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1 \therefore$ by mathematical induction the statement holds for all positive integral values of n .

Question 3

a) Prove by induction that for all positive integers n

$$\sum_{r=1}^n r \left(\frac{1}{2}\right)^{r-1} = 4 - \frac{n+2}{2^{n-1}}$$

b) Hence evaluate

$$\sum_{r=4}^{10} r \left(\frac{1}{2}\right)^{r-1}$$

Suggested Solution

a) When $n = 1$:

LHS

$$\begin{aligned}\sum_{r=1}^n r \left(\frac{1}{2}\right)^{r-1} &\equiv \sum_{r=1}^1 r \left(\frac{1}{2}\right)^{r-1} \\ &= (1) \left(\frac{1}{2}\right)^{1-1} = (1) \left(\frac{1}{2}\right)^0 \\ &= (1)(1) = 1\end{aligned}$$

RHS

$$\begin{aligned}4 - \frac{1+2}{2^{1-1}} &= 4 - \frac{3}{2^0} \\ &= 4 - \frac{3}{1} \\ &= 4 - 3 = 1\end{aligned}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\sum_{r=1}^k r \left(\frac{1}{2}\right)^{r-1} = 4 - \frac{k+2}{2^{k-1}}$$

Then for $n = k + 1$:

$$\sum_{r=1}^{k+1} r \left(\frac{1}{2}\right)^{r-1} = 4 - \frac{k+1+2}{2^{k+1-1}}$$

Proof

$$\sum_{r=1}^{k+1} r \left(\frac{1}{2}\right)^{r-1} = 4 - \frac{k+3}{2^k}$$

LHS

$$\begin{aligned} \sum_{r=1}^{k+1} r \left(\frac{1}{2}\right)^{r-1} &= \sum_{r=1}^k r \left(\frac{1}{2}\right)^{r-1} + (k+1)th \text{ term} \\ &= 4 - \frac{k+2}{2^{k-1}} + (k+1) \left(\frac{1}{2}\right)^{k+1-1} \\ &= 4 - \frac{k+2}{2^{k-1}} + (k+1) \left(\frac{1}{2}\right)^k \\ &= \frac{4(2^{k-1}) - (k+2)}{2^{k-1}} + (k+1) \left(\frac{1}{2}\right)^k \\ &= \frac{2(2^k) - k - 2}{2^{k-1}} + (k+1) \left(\frac{1}{2}\right)^k \\ &= \frac{2(2^k) - k - 2}{\frac{2^k}{2}} + (k+1) \left(\frac{1}{2}\right)^k \\ &= \frac{2[2(2^k) - k - 2]}{2^k} + (k+1) \left(\frac{1}{2}\right)^k \\ &= \frac{4(2^k) - 2k - 4 + k + 1}{2^k} \\ &= \frac{4(2^k) - k - 3}{2^k} \\ &= \frac{4(2^k) - (k+3)}{2^k} \\ &= \frac{4(2^k)}{2^k} - \frac{k+3}{2^k} \\ &= 4 - \frac{k+3}{2^k} \quad (\text{which is true for } n = k+1) \end{aligned}$$

∴ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for all positive integers n

$$\begin{aligned} \text{b) } \sum_{r=4}^{10} r \left(\frac{1}{2}\right)^{r-1} &= \sum_{r=1}^{10} r \left(\frac{1}{2}\right)^{r-1} - \sum_{r=1}^3 r \left(\frac{1}{2}\right)^{r-1} \\ &= \left(4 - \frac{10+2}{2^{10-1}}\right) - \left(4 - \frac{3+2}{2^{3-1}}\right) \\ &= \left(4 - \frac{12}{2^9}\right) - \left(4 - \frac{5}{2^2}\right) \\ &= \frac{5}{4} - \frac{12}{512} \\ &= \frac{157}{128} \text{ or } 1\frac{29}{128} \text{ or } 1.226\,562\,5 (= 1.2 \text{ to } 2 \text{ s.f.}) \end{aligned}$$

Question 4

ZIMSEC November 2016 Paper 2

Prove by induction that

$$\sum_{r=1}^n 4^r = \frac{4}{3}(4^n - 1),$$

for all positive integral values of n .

[7]

Suggested Solution

When $n = 1$:

LHS

$$\begin{aligned} \sum_{r=1}^n 4^r &\equiv \sum_{r=1}^1 4^r \\ &= (4)^1 = 4 \end{aligned}$$

RHS

$$\begin{aligned}\frac{4}{3}(4^n - 1) &= \frac{4}{3}(4^1 - 1) \\ &= \frac{4}{3} \times 3 = 4\end{aligned}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\sum_{r=1}^k 4^r = \frac{4}{3}(4^k - 1)$$

Then for $n = k + 1$:

$$\sum_{r=1}^{k+1} 4^r = \frac{4}{3}(4^{k+1} - 1)$$

Proof

$$\sum_{r=1}^{k+1} 4^r = \frac{4}{3}(4^{k+1} - 1)$$

LHS

$$\begin{aligned}\sum_{r=1}^{k+1} 4^r &= \sum_{r=1}^k 4^r + (k+1)th \text{ term} \\ &= \frac{4}{3}(4^k - 1) + 4^{k+1} \\ &= \frac{4}{3}(4^k) \frac{4}{3} + 4^{k+1} \\ &= \frac{4^{k+1}}{3} - \frac{4}{3} + 4^{k+1} \\ &= 4^{k+1} + \frac{4^{k+1}}{3} - \frac{4}{3}\end{aligned}$$

$$= 4^{k+1} \left(1 + \frac{1}{3} \right) - \frac{4}{3}$$

$$= 4^{k+1} \left(\frac{4}{3} \right) - \frac{4}{3}$$

$$= \frac{4}{3} (4^{k+1} - 1) \text{ (which is true for } n = k + 1 \text{)}$$

∴ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for all positive integers n

PRACTICE QUESTIONS

Question 1

Prove by induction that: $1 + 2 + 3 + \dots + n = \frac{n}{2}(n + 1)$ for any $n \geq 1$.

Question 2

Prove by induction that:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n + 1)(2n + 1) \text{ for all } n \in \mathbb{Z}^+$$

Question 3

Use mathematical induction to prove that:

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n + 1)^2 \text{ for } n \in \mathbb{Z}^+$$

Question 4

Use mathematical induction to prove that:

$$1^3 + 2^3 + 3^3 + \dots + r^3 = \frac{1}{4}n^2(n + 1)^2 \text{ for } n \in \mathbb{Z}^+.$$

Question 5

Prove by induction that for $n \geq 1$;

$$\sum_{r=1}^n (3r-2)^2 = \frac{1}{2}n(6n^2 - 3n - 1)$$

Question 6

Prove by induction that for $n \geq 1$;

$$1 + 4 + 7 + \dots + (3n-2)^2 = \frac{1}{2}n(6n^2 - 3n - 1)$$

Question 7

Prove by induction that:

$$\sum_{r=1}^n (2r-1) = n^2 \text{ for all positive integral values of } n.$$

Question 8

Prove by induction that:

$$\sum_{r=1}^n (3r-2) = \frac{n}{2}(3n-1) \text{ for all } n \in \mathbb{Z}^+$$

Question 9

a) Prove by induction that for all positive integers n

$$\sum_{r=1}^n 10^r = \frac{10}{9}(10^n - 1)$$

b) Hence evaluate

$$\sum_{r=3}^5 10^r$$

PAST EXAMINATION QUESTIONS

Question 1

ZIMSEC NOVEMBER 2008 PAPER 2

Prove by induction that:

$$\sum_{r=1}^n (r^3 - r) = \frac{1}{4}n(n-1)(n+1)(n+2) \text{ for all } n \in \mathbb{Z}^+$$

Hence evaluate

$$\sum_{r=3}^{50} (r^3 - r)$$

[8]

Question 2

ZIMSEC JUNE 2010 PAPER 2

Prove by mathematical induction that

$$\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \cdots + \frac{1}{4n^2 - 1} = \frac{n}{2n + 1}$$

[7]

Question 3

ZIMSEC NOVEMBER 2011 PAPER 2

Prove by induction that

$$\sum_{r=1}^n r(2^{r-1}) = 1 + (n-1)2^n$$

for all positive integral values of n .

[7]

Question 4

ZIMSEC JUNE 2012 PAPER 2

Prove by induction that

$$\sum_{r=1}^n (r+3)(2r+1) = \frac{n}{6}(4n^2 + 27n + 41)$$

[7]

Question 5

ZIMSEC JUNE 2015 PAPER 2

Prove by induction that

$$\sum_{r=1}^n a p^r = ap \left[\frac{1 - p^n}{1 - p} \right]$$

for all positive integral values of n , where a and p are constants.

[7]

Question 6

ZIMSEC NOVEMBER 2015 PAPER 2

Prove by mathematical induction that

$$2(2^1) + 3(2^2) + 4(2^3) + \dots + n(2^{n-1}) = 4(n-1)(2^{n-2})$$

for all natural numbers greater than 1.

[7]

Question 7

ZIMSEC JUNE 2017 PAPER 2

Prove by induction that:

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1)$$

[7]

Question 8

ZIMSEC NOVEMBER 2017 PAPER 2

Prove by induction that

$$\sum_{r=1}^n (r+1) 2^r = n2^{n+1}$$

for all $n \in \mathbb{Z}^+$, where $\mathbb{Z}^+$ is the set of positive integers.

[8]

Question 9

ZIMSEC JUNE 2019 PAPER 2

Prove by induction that:

$$\sum_{r=1}^n r(r+1) = \frac{1}{3}n(n+1)(n+2)$$

for all positive integers n .

[7]

Question 10

ZIMSEC NOVEMBER 2019 PAPER 2

Prove by induction the statement

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1} : r \neq 1$$

[7]

MATRICES

PROOF HINT

Suppose we want to prove that $A^n = A$

(i) Assumption $A^k = A$

(ii) Proof $A^{k+1} = A$

Apply the laws of indices to the LHS and substitute the Assumption i.e.

$$A^{k+1} = A^k \times A$$

$\therefore$ Premultiply the two matrices with assumption i.e. A^k

Examples

Question'1

Prove by mathematical induction that if $Z = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$, then

$$Z^n = \begin{pmatrix} n+1 & -n \\ n & 1-n \end{pmatrix} \text{ for all } n \in \mathbb{Z}^+.$$

Suggested Solution

When $n = 1$:

$$LHS = Z^1 = Z = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$$

$$RHS = \begin{pmatrix} 1+1 & -1 \\ 1 & 1-1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \equiv LHS$$

Since $RHS \equiv LHS = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$Z^k = \begin{pmatrix} k+1 & -k \\ k & 1-k \end{pmatrix}$$

Then for $n = k + 1$:

$$Z^{k+1} = \begin{pmatrix} [(k+1)+1] & -(k+1) \\ k+1 & 1-(k+1) \end{pmatrix}$$

Proof

Note:

$$Z^{k+1} = Z^k \times Z^1$$

$$Z^{k+1} = \begin{pmatrix} [(k+1)+1] & -(k+1) \\ k+1 & 1-(k+1) \end{pmatrix}$$

LHS:

$$\begin{aligned} Z^{k+1} &= Z^k \cdot Z^1 = \begin{pmatrix} k+1 & -k \\ k & 1-k \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 2(k+1) - k & -1(k+1) + 0 \\ 2k + 1(1-k) & -k + 0 \end{pmatrix} \\ &= \begin{pmatrix} 2k + 2 - k & -k - 1 \\ 2k + 1 - k & -k \end{pmatrix} \\ &= \begin{pmatrix} k + 2 & -(k+1) \\ k + 1 & -k + 1 - 1 \end{pmatrix} \\ &= \begin{pmatrix} k + 2 & -(k+1) \\ k + 1 & 1 - (k+1) \end{pmatrix} \text{ which is true for } n = k + 1. \end{aligned}$$

$\therefore$ The statement is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1 \therefore$ by mathematical induction the statement is also true for $n \in \mathbb{Z}^+$

Question'2

If $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, prove by induction to show that $A^n = \begin{pmatrix} 1 & na \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{N}$.

Suggested Solution

When $n = 1$:

LHS

$$A^n = A^1$$

$$= A$$

$$= \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

RHS

$$\begin{pmatrix} 1 & na \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \times a \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$A^k = \begin{pmatrix} 1 & ka \\ 0 & 1 \end{pmatrix}$$

Then for $n = k + 1$:

$$A^{k+1} = \begin{bmatrix} 1 & (k+1)a \\ 0 & 1 \end{bmatrix}$$

Proof

$$A^{k+1} = \begin{bmatrix} 1 & (k+1)a \\ 0 & 1 \end{bmatrix}$$

LHS

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 1 & ka \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned}
&= \begin{pmatrix} 1 \times 1 + ka \times 0 & 1 \times a + ka \times 1 \\ 0 \times 1 + 1 \times 0 & 0 \times a + 1 \times 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & a + ka \\ 0 & 1 \end{pmatrix} \\
&= \begin{bmatrix} 1 & (k+1)a \\ 0 & 1 \end{bmatrix} \text{ (which is true for } n = k+1 \text{)}
\end{aligned}$$

$\therefore$ The statement holds for $n = k+1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k+1$ then by induction the statement is true for all $n \in \mathbb{N}$.

TROCKERS

PRACTICE QUESTIONS

Question 1

Prove by induction that $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}^{2n+1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$.

Question 2

Prove by mathematical induction that if $Q = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, then $Q^n = \begin{pmatrix} 1 & n & \frac{n}{2}(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

for all $n \in \mathbb{Z}^+$.

Question 3

Prove by mathematical induction that if $A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$, then $A^n = \begin{pmatrix} 1 & 3n \\ 0 & 1 \end{pmatrix}$

for all $n \in \mathbb{Z}^+$.

Question 4

Prove by mathematical induction that if $B = \begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$, then $B^n = \begin{pmatrix} 2n+1 & -n \\ 4n & 1-2n \end{pmatrix}$

for all $n \in \mathbb{Z}^+$.

Question 5

Prove by mathematical induction that if $B = \begin{pmatrix} 1 + \cos\theta & \sin\theta \\ \sin\theta & 1 - \cos\theta \end{pmatrix}$, then

$B^{n+1} = 2^n \times B$ for all $n \in \mathbb{Z}^+$.

Question 6

Prove by mathematical induction that $\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}^n = \begin{pmatrix} 1 & 1 - 2^n \\ 0 & 2^n \end{pmatrix}$ for $n \in \mathbb{Z}^+$.

Question 7

Prove by mathematical induction that $\begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}^n = \begin{pmatrix} -3n + 1 & 9n \\ -n & 3n + 1 \end{pmatrix}$ for $n \in \mathbb{Z}^+$.

Question 8

Prove by mathematical induction that if $P = \begin{pmatrix} 2 & 0 \\ -1 & 1 \end{pmatrix}$, then $P^n = \begin{pmatrix} 2^n & 0 \\ 1 - 2^n & 1 \end{pmatrix}$ for $n \in \mathbb{Z}^+$.

Question 9

If $A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, then prove that $A^n = \begin{pmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{pmatrix}$ for $n \in \mathbb{Z}^+$.

Question 10

Prove by mathematical induction that $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n + 1 & -4n \\ n & -2n + 1 \end{pmatrix}$ for $n \in \mathbb{Z}^+$.

PAST EXAMINATION QUESTION

Question 1

ZIMSEC JUNE 2004 PAPER 2

A transformation matrix M is given by $M = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$

a) Show that, under this transformation

(i) all points on the line $y = x$ are invariant points; [2]

(ii) any point on the, line $y = x + c$ where $c \neq 0$, is transformed to another point on that line. [2]

b) Prove by induction that, if n is an integer, then $M^n = \begin{pmatrix} n+1 & -n \\ n & 1-n \end{pmatrix}$. [5]

Question 2

ZIMSEC NOVEMBER 2004 PAPER 2

Prove by Mathematical Induction that

$$\begin{pmatrix} 1 & a \\ 0 & a \end{pmatrix}^n = \begin{pmatrix} 1 & \frac{a^{n+1} - a}{a - 1} \\ 0 & a^n \end{pmatrix} \text{ for all } n \in \mathbb{Z}^+. \quad [7]$$

Hence simplify the matrix $\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}^6$. [2]

Question 3

ZIMSEC JUNE 2007 PAPER 2

Prove by Mathematical Induction that $\begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}^{2n} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, where n is a positive integer.

[6]

Question 4

ZIMSEC NOVEMBER 2012 PAPER 2

If $A = \begin{pmatrix} 2 & a \\ 0 & 1 \end{pmatrix}$, prove by induction that for every positive integer n ,

$$A^n = \begin{pmatrix} 2^n & (2^n - 1)a \\ 0 & 1 \end{pmatrix} \quad [7]$$

TROCKERS

INTEGRATION

PROOF HINT

Suppose we want to prove that $\int A^n dx = B$

(i) Assumption $\int A^k dx = B$

(ii) Proof $\int A^{k+1} dx = B$

Apply the laws of indices to the LHS and use integration by parts but you must substitute the Assumption in your proof i.e.

$$\int A^{k+1} = \int A^k \cdot A dx$$

Examples

Question 1

Use induction to prove that

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \text{ for } n \in \mathbb{Z}^+.$$

Suggested Solution

When $n = 1$:

$$LHS = \int x^1 dx = \frac{x^2}{2} + c$$

$$RHS = \frac{x^{1+1}}{1+1} + c = \frac{x^2}{2} + c \equiv LHS$$

Since $RHS \equiv LHS = \frac{x^2}{2} + c \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$

$$\int x^k = \frac{x^{k+1}}{k+1} + c$$

Then for $n = k + 1$

$$\int x^{k+1} = \frac{x^{(k+1)+1}}{(k+1)+1} + c$$

Proof

Note:

(i) $\int x^{k+1} dx \equiv \int x^k \cdot x^1 dx$. Then use the method of integration by parts. Choose u to be an expression which vanishes after performing 1 or more derivations.

(ii) To integrate $\frac{dv}{dx}$ i.e. $\int x^k dx$ use the assumption stage result.

$$\int x^{k+1} = \frac{x^{(k+1)+1}}{(k+1)+1} + c$$

LHS:

$$\int x^{k+1} dx = \int x^k \cdot x^1 dx$$

$$\text{Let } u = x^1 \Rightarrow \frac{du}{dx} = 1 \text{ and}$$

$$\frac{dv}{dx} = x^k \Rightarrow \int \frac{dv}{dx} dx = \int x^k dx$$

$$\Rightarrow v = \frac{x^{k+1}}{k+1} \text{ (from the assumption stage)}$$

Now use

$$\int u \frac{dv}{dx} = uv - \int v \frac{du}{dx} dx$$

$$\begin{aligned} \int x^{k+1} dx &= x \frac{x^{k+1}}{k+1} - \int \frac{x^{k+1}}{k+1} (1) dx \\ &= x \cdot \frac{x^{k+1}}{k+1} - \frac{x^{(k+1)+1}}{(k+1)[(k+1)+1]} + c \end{aligned}$$

$$\begin{aligned}
&= \frac{x^{(k+1)+1}}{k+1} - \frac{x^{(k+1)+1}}{(k+1)[(k+1)+1]} + c \text{ (Since } a^b \times a^c = a^{b+c}\text{)} \\
&= \frac{x^{(k+1)+1}}{k+1} \left\{ 1 - \frac{1}{[(k+1)+1]} \right\} + c \\
&= \frac{x^{(k+1)+1}}{k+1} \left\{ \frac{[(k+1)+1]}{[(k+1)+1]} - \frac{1}{[(k+1)+1]} \right\} + c \\
&= \frac{x^{(k+1)+1}}{k+1} \left\{ \frac{[(k+1)+1] - 1}{[(k+1)+1]} \right\} + c \\
&= \frac{x^{(k+1)+1}}{k+1} \left[\frac{k+1}{[(k+1)+1]} \right] + c \\
&= \frac{x^{(k+1)+1}}{(k+1)+1} + c \text{ (which is true for } n = k+1\text{)}
\end{aligned}$$

∴ The statement is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1$ ∴ by mathematical induction the statement holds for $n \in \mathbb{Z}^+$.

Question 2

Prove that for all $n \geq 0$,

$$\int_0^{\infty} x^n e^{-x} dx = n!$$

Suggested Solution

When $n = 0$:

LHS:

$$\begin{aligned}
\int_0^{\infty} x^n e^{-x} dx &= \int_0^{\infty} x^0 e^{-x} dx \\
&= \int_0^{\infty} e^{-x} dx
\end{aligned}$$

$$\begin{aligned}
 &= \left[\frac{e^{-x}}{-1} \right]_0^{\infty} \\
 &= [-e^{-x}]_0^{\infty} \\
 &= (-e^{-\infty}) - (-e^0) \\
 &= 1
 \end{aligned}$$

RHS:

$$n! = 0! = 1$$

Since LHS = RHS $\therefore$ the statement is true for $n = 0$.

Assuming true for $n = k$:

$$\int_0^{\infty} x^k e^{-x} dx = k!$$

Then for $n = k + 1$:

$$\int_0^{\infty} x^{k+1} e^{-x} dx = (k + 1)!$$

Proof

$$\int_0^{\infty} x^{k+1} e^{-x} dx = (k + 1)!$$

LHS

$$\int_0^{\infty} x^{k+1} e^{-x} dx$$

Let $u = x^{k+1}$

$$\Rightarrow \frac{du}{dx} = (k + 1)x^k$$

$$\frac{dv}{dx} = e^{-x} \Rightarrow v = -e^{-x}$$

Now:

$$\begin{aligned}\int_0^{\infty} x^{k+1} e^{-x} dx &= uv - \int_0^{\infty} v \left(\frac{du}{dx} \right) dx \\&= [-e^{-x} x^{k+1}]_0^{\infty} - \int_0^{\infty} (-e^{-x})(k+1)x^k dx \\&= [-e^{-x} x^{k+1}]_0^{\infty} + (k+1) \int_0^{\infty} (e^{-x}) x^k dx \\&= [-e^{-x} x^{k+1}]_0^{\infty} + (k+1)(k!) \text{ (from the assumption)} \\&= \{(-e^{-\infty} \infty^{k+1}) - (-e^{-\infty} 0^{k+1})\} + (k+1)(k!)\end{aligned}$$

Remember that $e^{-\infty} = 0$

$$= 0 + (k+1)(k!)$$

Remember that $(k+1)(k!) = (k+1)!$

$$= (k+1)! \text{ (which is true for } n = k+1 \text{)}$$

$\therefore$ The statement holds for $n = k+1$.

Since the statement holds for $n = 0$; $n = k$ and $n = k+1$ then by induction the statement is true for $n \geq 0$.

PRACTICE QUESTIONS

Question 1

Use induction to prove that if m is a constant

$$\int mx^{n+1}dx = \frac{mx^{n+2}}{n+2} + c \text{ for } n \in \mathbb{Z}^+.$$

Question 2

Use induction to prove that

$$\int_0^{\frac{\pi}{2}} \sin^{2n}x dx = \frac{(2n)! \pi}{(n!)^2 2^{2n+1}} \text{ for } n \geq 0.$$

Question 3

Use induction to prove that

$$\int x^n e^x dx = e^x \left[x^n - \frac{d(x^n)}{dx} + \frac{d^2(x^n)}{dx^2} + \dots + (-1)^n \frac{d^n(x^n)}{dx^n} \right] \text{ for } n \in \mathbb{N}$$

DIFFERENTIATION

PROOF HINT

Suppose we want to prove that $\frac{d^n(x)}{dx^n} = b$

(i) Assumption $\frac{d^k(x)}{dx^k} = b$

(ii) Proof $\frac{d^{k+1}(x)}{dx^{k+1}} = b$

Apply the laws of indices to the LHS and differentiate the Assumption i.e.

$$\frac{d^{k+1}(x)}{dx^{k+1}} = \frac{d}{dx} \left[\frac{d^k(x)}{dx^k} \right]$$

Examples

Question 1

Prove by induction that for $n \geq 1$

$$\frac{d(x^n)}{dx} = nx^{n-1}$$

Suggested Solution

When $n = 1$:

$$LHS = \frac{d(x^1)}{dx} = 1$$

$$RHS = 1 \cdot x^{1-1} = 1 \cdot x^0 = 1 \cdot 1 = 1 \equiv LHS$$

Since $RHS \equiv LHS = 1 \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$

$$\frac{d(x^k)}{dx} = kx^{k-1}$$

Then for $n = k + 1$:

$$\frac{d(x^{k+1})}{dx} = (k + 1)x^{(k+1)-1}$$

Proof

Note:

$$\frac{d(x^{k+1})}{dx} \equiv \frac{d(x^k \cdot x^1)}{dx} : \text{Then use the product rule (differentiate the assumption)}$$

$$\frac{d(x^{k+1})}{dx} = (k + 1)x^{(k+1)-1}$$

LHS:

$$\frac{d(x^{k+1})}{dx} = \frac{d(x^k \cdot x^1)}{dx}$$

$$\text{Let } v = x^1 \Rightarrow \frac{dv}{dx} = 1 \text{ and}$$

$$u = x^k \Rightarrow \frac{du}{dx} = kx^{k-1} \text{ (from the assumption stage)}$$

$$\text{Now use } \frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d(x^{k+1})}{dx} = x^k(1) + x^1(kx^{k-1})$$

$$= x^k(1) + kx^k$$

$$= x^k(k + 1)$$

$$= (k + 1)x^k$$

$$= (k + 1)x^{k+1-1}$$

$$= (k + 1)x^{(k+1)-1} \text{ (which is true for } n = k + 1 \text{)}$$

∴ The statement is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1$ ∴ by mathematical induction the statement must hold for $n \in \mathbb{Z}^+$.

Question 2

Use Mathematical Induction to prove that if $y = xe^x$ then

$$\frac{d^n y}{dx^n} = (x + n)e^x$$

Suggested Solution

When $n = 1$:

$$LHS = \frac{d^1 y}{dx^1} = \frac{dy}{dx} = xe^x + e^x (1) = xe^x + e^x = (x + 1)e^x$$

$$RHS = (x + 1)e^x \equiv LHS$$

Since $RHS \equiv LHS = 1$ ∴ the statement is true for $n = 1$.

Assuming that the statement is true for $n = k$:

$$\frac{d^k y}{dx^k} = (x + k)e^x$$

Then for $n = k + 1$:

$$\frac{d^{k+1} y}{dx^{k+1}} = [x + (k + 1)]e^x$$

Proof

Note:

$$\frac{d^{k+1}y}{dx^{k+1}} \equiv \frac{d^{k+1}y}{dx^{k+1}} \equiv \frac{d^k}{dx^k} \left[\frac{d(y)}{dx} \right] \equiv \frac{d}{dx} \left(\frac{d^k y}{dx^k} \right) : \text{Differentiate the assumption result}$$

$$\frac{d^{k+1}y}{dx^{k+1}} = \frac{d^k}{dx^k} \left[\frac{d(y)}{dx} \right]$$

$$= \frac{d}{dx} \left(\frac{d^k}{dx^k} \right)$$

$$= \frac{d}{dx} [(x+k)e^x]$$

$$= (x+k)e^x + e^x(1)$$

$$= e^x [x + (k+1)]$$

$$= [x + (k+1)]e^x \quad (\text{which is true for } n = k+1)$$

∴ The statement is true for $n = k+1$.

Since the statement is true for $n = 1, n = k, n = k+1$ ∴ by mathematical induction the statement holds for $n \in \mathbb{Z}^+$.

Question 3

Given that a is a constant, prove by induction that, for every positive integer n ,

$$\frac{d^n}{dx^n} (xe^{ax}) = na^{n-1}e^{ax} + a^n xe^{ax}$$

Suggested Solution

When $n = 1$:

LHS

$$\frac{d^n}{dx^n} (xe^{ax}) = x(ae^{ax}) + e^{ax}(1) = axe^{ax} + e^{ax}$$

RHS

$$na^{n-1}e^{ax} + a^n xe^{ax} = (1)a^{1-1}e^{ax} + a^{(1)}xe^{ax} = e^{ax} + axe^{ax}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\frac{d^k}{dx^k}(xe^{ax}) = ka^{k-1}e^{ax} + a^kxe^{ax}$$

Then for $n = k + 1$:

$$\frac{d^{k+1}}{dx^{k+1}}(xe^{ax}) = (k+1)a^{(k+1)-1}e^{ax} + a^{k+1}xe^{ax}$$

Proof

$$\frac{d^{k+1}}{dx^{k+1}}(xe^{ax}) = (k+1)a^ke^{ax} + a^{k+1}xe^{ax}$$

LHS

$$\begin{aligned}\frac{d^{k+1}}{dx^{k+1}}(xe^{ax}) &= \frac{d}{dx} \left[\frac{d^k}{dx^k}(xe^{ax}) \right] \\ &= \frac{d}{dx} (ka^{k-1}e^{ax} + a^kxe^{ax}) \text{ [from the assumption]} \\ &= a(ka^{k-1}e^{ax}) + [a^kx(ae^{ax}) + e^{ax}(a^k)] \\ &= [k(a^{k-1} \times a)e^{ax}] + [(a^k \times a)x(e^{ax}) + a^ke^{ax}] \\ &= [k(a^{k+1-1})e^{ax}] + [(a^{k+1})x(e^{ax}) + a^ke^{ax}] \\ &= ka^ke^{ax} + a^{k+1}xe^{ax} + a^ke^{ax} \\ &= ka^ke^{ax} + a^ke^{ax} + a^{k+1}xe^{ax} \\ &= (k+1)a^ke^{ax} + a^{k+1}xe^{ax} \text{ (which is true for } n = k + 1\text{)}\end{aligned}$$

$\therefore$ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for every positive integer n .

Question 4

Show that $\sin\left(x + \frac{\pi}{2}\right) \equiv \cos x$.

Hence prove by induction that

$$\frac{d^n}{dx^n}(\sin ax) = a^n \sin\left(ax + \frac{n\pi}{2}\right)$$

Suggested Solution

$$\sin\left(x + \frac{\pi}{2}\right) \equiv \cos x$$

LHS:

$$\begin{aligned}\sin\left(x + \frac{\pi}{2}\right) &\equiv \sin x \cos \frac{\pi}{2} + \cos x \sin \frac{\pi}{2} \\ &= \sin x(0) + \cos x(1) \\ &= 0 + \cos x \\ &= \cos x \equiv \text{RHS (as required)}\end{aligned}$$

Proving that

$$\frac{d^n}{dx^n}(\sin ax) = a^n \sin\left(ax + \frac{n\pi}{2}\right)$$

When $n = 1$:

LHS

$$\frac{d^1}{dx^1}(\sin ax) = \frac{d}{dx}(\sin ax) = a \cos ax$$

RHS

$$\begin{aligned}a^n \sin\left(ax + \frac{\pi}{2}\right) &\equiv a^1 \left[\sin(ax) \cos\left(\frac{\pi}{2}\right) + \cos(ax) \sin\left(\frac{\pi}{2}\right) \right] \\ &= a[\sin(ax)(0) + \cos(ax)(1)] \\ &= a[0 + \cos(ax)] \\ &= a \cos ax\end{aligned}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\frac{d^k}{dx^k}(\sin ax) = a^k \sin\left(ax + \frac{k\pi}{2}\right)$$

Then for $n = k + 1$:

$$\frac{d^{k+1}}{dx^{k+1}}(\sin ax) = a^{k+1} \sin\left[ax + \frac{(k+1)\pi}{2}\right]$$

Proof

$$\frac{d^{k+1}}{dx^{k+1}}(\sin ax) = a^{k+1} \sin\left[ax + \frac{(k+1)\pi}{2}\right]$$

LHS

$$\begin{aligned}\frac{d^{k+1}}{dx^{k+1}}(\sin ax) &= \frac{d}{dx} \left[\frac{d^k}{dx^k}(\sin ax) \right] \\ &= \frac{d}{dx} \left[a^k \sin\left(ax + \frac{k\pi}{2}\right) \right] \text{ [from the assumption]} \\ &= a^k \cos\left(ax + \frac{k\pi}{2}\right) \times a \\ &= a^k \times a \times \cos\left(ax + \frac{k\pi}{2}\right) \\ &= a^{k+1} \cos\left(ax + \frac{k\pi}{2}\right)\end{aligned}$$

NB:

$$\text{Let } X = ax + \frac{k\pi}{2}$$

$$\Rightarrow \frac{d^{k+1}}{dx^{k+1}}(\sin ax) = a^{k+1} \cos(X)$$

But

$$\sin\left(x + \frac{\pi}{2}\right) \equiv \cos x$$

$$\Rightarrow \frac{d^{k+1}}{dx^{k+1}}(\sin ax) = a^{k+1} \sin\left(X + \frac{\pi}{2}\right)$$

$$\text{Remember } X = ax + \frac{k\pi}{2}$$

$$\begin{aligned} \Rightarrow \frac{d^{k+1}}{dx^{k+1}}(\sin ax) &= a^{k+1} \sin\left(ax + \frac{k\pi}{2} + \frac{\pi}{2}\right) \\ &= a^{k+1} \sin\left[ax + \frac{(k+1)\pi}{2}\right] \text{ (which is true for } n = k+1) \end{aligned}$$

$\therefore$ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for every positive integer n .

Question 5

Show that $\cos\left(x + \frac{\pi}{2}\right) \equiv -\sin x$.

Hence prove by mathematical induction that for all positive integral values of n ,

$$\frac{d^n}{dx^n}[\cos(ax + b)] = a^n \cos\left(ax + b + \frac{n\pi}{2}\right)$$

Suggested Solution

$$\cos\left(x + \frac{\pi}{2}\right) \equiv -\sin x$$

LHS:

$$\begin{aligned} \cos\left(x + \frac{\pi}{2}\right) &\equiv \cos x \cos \frac{\pi}{2} - \sin x \sin \frac{\pi}{2} \\ &= \sin x(0) - \sin x(1) = 0 - \sin x \\ &= -\sin x \equiv \text{RHS (as required)} \end{aligned}$$

Proving that

$$\frac{d^n}{dx^n} [\cos(ax + b)] = a^n \cos\left(ax + b + \frac{n\pi}{2}\right)$$

When $n = 1$:

LHS

$$\frac{d^1}{dx^1} [\cos(ax + b)] = \frac{d}{dx} [\cos(ax + b)] = -a \sin(ax + b)$$

RHS

$$\begin{aligned} a^n \cos\left(ax + b + \frac{1}{2}n\pi\right) &\equiv a^1 \cos\left[ax + b + \frac{(1)\pi}{2}\right] \\ &= a \left[\cos(ax + b) \cos\left(\frac{\pi}{2}\right) - \sin(ax + b) \sin\left(\frac{\pi}{2}\right) \right] \\ &= a [\cos(ax + b)(0) - \sin(ax + b)(1)] \\ &= a [0 - \sin(ax + b)] \\ &= -a \sin(ax + b) \end{aligned}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\frac{d^k}{dx^k} [\cos(ax + b)] = a^k \cos\left(ax + b + \frac{k\pi}{2}\right)$$

Then for $n = k + 1$:

$$\frac{d^{k+1}}{dx^{k+1}} [\cos(ax + b)] = a^{k+1} \cos\left[ax + b + \frac{(k+1)k\pi}{2}\right]$$

Proof

$$\frac{d^{k+1}}{dx^{k+1}} [\cos(ax + b)] = a^{k+1} \cos\left[ax + b + \frac{(k+1)k\pi}{2}\right]$$

LHS

$$\frac{d^{k+1}}{dx^{k+1}} [\cos(ax + b)] = \frac{d}{dx} \left[\frac{d^k}{dx^k} [\cos(ax + b)] \right]$$

$$\begin{aligned}
&= \frac{d}{dx} \left[a^k \cos \left(ax + b + \frac{k\pi}{2} \right) \right] \text{ [from the assumption]} \\
&= -a^k \sin \left(ax + b + \frac{k\pi}{2} \right) \times a \\
&= -a^k \times a \times \sin \left(ax + b + \frac{k\pi}{2} \right) \\
&= -a^{k+1} \sin \left(ax + b + \frac{k\pi}{2} \right)
\end{aligned}$$

Aside:

$$\text{Let } X = ax + b + \frac{k\pi}{2}$$

$$\Rightarrow \frac{d^{k+1}}{dx^{k+1}} [\cos(ax + b)] = -a^{k+1} \sin(X)$$

But

$$\cos \left(x + \frac{\pi}{2} \right) \equiv -\sin x$$

$$\Rightarrow \frac{d^{k+1}}{dx^{k+1}} [\cos(ax + b)] = a^{k+1} \cos \left(X + \frac{\pi}{2} \right)$$

$$\text{Remember } X = ax + b + \frac{k\pi}{2}$$

$$\begin{aligned}
\Rightarrow \frac{d^{k+1}}{dx^{k+1}} [\cos(ax + b)] &= a^{k+1} \cos \left(ax + b + \frac{k\pi}{2} + \frac{\pi}{2} \right) \\
&= a^{k+1} \cos \left[ax + b + \frac{(k+1)\pi}{2} \right] \text{ (which is true for } n = k + 1)
\end{aligned}$$

$\therefore$ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for all positive integral values of n .

Question 6

(a) If $y = xe^x$, find:

(i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

(iii) $\frac{d^3y}{dx^3}$

(b) Suggest a formula for $\frac{d^ny}{dx^n}$

(c) Prove your result in (iv) by Mathematical Induction.

Suggested Solution

a) i) $\frac{dy}{dx} = xe^x + e^x (1) = xe^x + e^x = (x+1)e^x$ (Use the Product Rule)

ii) $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = [xe^x + e^x (1)] + e^x$
 $= xe^x + e^x + e^x$
 $= (x+2)e^x$

iii) $\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = [xe^x + e^x (1)] + e^x + e^x$
 $= xe^x + e^x + e^x + e^x$
 $= (x+3)e^x$

b) $\frac{d^ny}{dx^n} = (x+n)e^x$

c) Check example 2 for the proof.

Question 7

UCLES NOVEMBER 1992 PAPER 1

The variable x and y are related by $\frac{dy}{dx} = 2xy - 1$, and $y = 1$ at $x = 0$.

(a) Show that at $x = 0$, $\frac{d^3y}{dx^3} = -4$ and $\frac{d^4y}{dx^4} = 12$.

(b) Prove by induction that, for all values of the positive integer n ,

$$\frac{d^{n+2}y}{dx^{n+2}} = 2(n+1) \frac{d^ny}{dx^n} + 2x \frac{d^{n+1}y}{dx^{n+1}}$$

Suggested Solution

(a) $\frac{dy}{dx}_{x=0} = 2(0)y - 1 = -1$

$$\frac{dy}{dx} = 2xy - 1$$

$$\frac{d^2y}{dx^2} = 2x \frac{dy}{dx} + 2y \quad (\text{Use Implicit differentiation})$$

$$\frac{d^2y}{dx^2}_{x=0} = 2(0)(-1) + 2(1) = 2$$

Differentiate $\frac{d^2y}{dx^2}$ to get $\frac{d^3y}{dx^3}$.

$$\frac{d^3y}{dx^3} = 2x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + 2 \frac{dy}{dx}$$

$$\Rightarrow \frac{d^3y}{dx^3} = 2x \frac{d^2y}{dx^2} + 4 \frac{dy}{dx}$$

Now:

$$\begin{aligned} \frac{d^3y}{dx^3}_{x=0} &= 2(0)(2) + 4(-1) \\ &= -4 \quad (\text{As required}) \end{aligned}$$

Differentiate $\frac{d^3y}{dx^3}$ to get $\frac{d^4y}{dx^4}$.

$$\frac{d^4y}{dx^4} = 2x \frac{d^3y}{dx^3} + 2 \frac{d^2y}{dx^2} + 4 \frac{d^2y}{dx^2}$$

$$\Rightarrow \frac{d^4y}{dx^4} = 2x \frac{d^3y}{dx^3} + 6 \frac{d^2y}{dx^2}$$

Now:

$$\frac{d^4 y}{dx^4} \Big|_{x=0} = 2(0)(-4) + 6(2)$$

$$= 12 \text{ (As required)}$$

(b) When $n = 1$

LHS:

$$\begin{aligned} \frac{d^{n+2} y}{dx^{n+2}} &= \frac{d^{1+2} y}{dx^{1+2}} = \frac{d^3 y}{dx^3} \\ &= 2x \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} \end{aligned}$$

LHS:

$$\begin{aligned} 2(n+1) \frac{d^n y}{dx^n} + 2x \frac{d^{n+1} y}{dx^{n+1}} &= 2(1+1) \frac{d^1 y}{dx^1} + 2x \frac{d^{1+1} y}{dx^{1+1}} \\ &= 2(2) \frac{dy}{dx} + 2x \frac{d^2 y}{dx^2} \\ &= 2x \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} \end{aligned}$$

Since $LHS = RHS \therefore$ the statement holds for $n = 1$.

Assuming true for $n = k$:

$$\frac{d^{k+2} y}{dx^{k+2}} = 2(k+1) \frac{d^k y}{dx^k} + 2x \frac{d^{k+1} y}{dx^{k+1}}$$

Then for $n = k + 1$:

$$\frac{d^{k+1+2} y}{dx^{k+1+2}} = 2(k+1+1) \frac{d^{k+1} y}{dx^{k+1}} + 2x \frac{d^{k+1+1} y}{dx^{k+1+1}}$$

Proof:

$$\frac{d^{k+3} y}{dx^{k+3}} = 2(k+2) \frac{d^{k+1} y}{dx^{k+1}} + 2x \frac{d^{k+2} y}{dx^{k+2}}$$

LHS:

$$\frac{d^{k+3} y}{dx^{k+3}} = \frac{d}{dx} \left(\frac{d^{k+2} y}{dx^{k+2}} \right) \quad (\text{Differentiate the assumption or Hypothesis stage})$$

$$= \frac{d}{dx} \left(2(k+1) \frac{d^k y}{dx^k} + 2x \frac{d^{k+1} y}{dx^{k+1}} \right)$$

$$\text{Apply the product rule for } \left(2x \frac{d^{k+1} y}{dx^{k+1}} \right)$$

$$\begin{aligned}
&= 2(k+1) \frac{d^{k+1}y}{dx^{k+1}} + 2x \frac{d^{k+2}y}{dx^{k+2}} + 2 \frac{d^{k+1}y}{dx^{k+1}} \\
&= 2(k+1) \frac{d^{k+1}y}{dx^{k+1}} + 2 \frac{d^{k+1}y}{dx^{k+1}} + 2x \frac{d^{k+2}y}{dx^{k+2}} \\
&= 2(k+2) \frac{d^{k+1}y}{dx^{k+1}} + 2x \frac{d^{k+2}y}{dx^{k+2}} \text{ (Which is true for } n = k+1 \text{)}
\end{aligned}$$

∴ The statement holds for $n = k + 1$.

Since the statement is true for $n = 1, n = k$ and $n = k + 1$ ∴ by induction the statement is true for all values of the positive integer n .

PRACTICE QUESTIONS

Question 1

Use mathematical induction to prove that if $y = \frac{1}{2x+1}$ then

$$\frac{d^n y}{dx^n} = \frac{(-2)^n n!}{(2x+1)^{n+1}}$$

Question 2

Show that $\cos\left(x + \frac{\pi}{2}\right) \equiv -\sin x$.

Hence prove by induction that

$$\frac{d^n}{dx^n}(\cos ax) = a^n \cos\left(ax + \frac{n\pi}{2}\right)$$

Question 3

If $p = \frac{1}{x+5}$, then prove by mathematical induction that for $n \geq 1$:

$$\frac{d^n p}{dx^n} = \frac{(-1)^n n!}{(x+5)^{n+1}}$$

Question 4

If $\frac{dy}{dx} = 2xy - 1$, show by induction that

$$\frac{d^{n+2}y}{dx^{n+2}} = 2(n+1) \frac{d^ny}{dx^n} + 2x \frac{d^{n+1}y}{dx^{n+1}}.$$

Question 5

Prove by Induction that

$$\frac{d^n}{dx^n}(e^x \cos x) = 2^{\frac{n}{2}} e^x \cos\left(x + \frac{n\pi}{4}\right) \text{ for } n \geq 1$$

Question 6

Show that $\cos\left(x + \frac{\pi}{2}\right) \equiv -\sin x$.

Hence prove by mathematical induction that for all positive integral values of n ,

$$\frac{d^n}{dx^n}[\sin(ax + b)] = a^n \sin\left(ax + b + \frac{n\pi}{2}\right)$$

Question 7

Prove by Induction that if $y = x^{-m}$ for $n \geq 1$

$$\frac{d^ny}{dx^n} = (-1)^n \frac{(m+n-1)!}{(m-1)!} \frac{1}{x^{m+n}}; n \geq m$$

Question 8

Prove by Induction that if $y = (ax + b)^{-m}$ for $n \geq 1$

$$\frac{d^ny}{dx^n} = (-1)^n \frac{(m+n-1)!}{(m-1)!} \frac{a^n}{(ax + b)^{m+n}}; n \geq m$$

Question 9

Prove by Induction that if $y = e^{ax}$ for $n \geq 1$

$$\frac{d^n y}{dx^n} = a^n e^{ax}$$

Question 10

Prove by Induction that if $y = e^{(ax+b)}$ for $n \geq 1$

$$\frac{d^n y}{dx^n} = a^n e^{(ax+b)}$$

Question 11

(a) If $y = -4e^{-2x}$, find:

(i) $\frac{dy}{dx}$

(ii) $\frac{d^2 y}{dx^2}$

(iii) $\frac{d^3 y}{dx^3}$

(iv) $\frac{d^4 y}{dx^4}$

(b) Suggest a formula for $\frac{d^n y}{dx^n}$

(c) Prove your formula by mathematical induction.

PAST EXAMINATION QUESTION

Question 1

ZIMSEC JUNE 2016 PAPER 2

It is given that $y = x^2e^x$,

(i) Show that $\frac{d^3y}{dx^3} = 4e^x + 2xe^x + \frac{d^2y}{dx^2}$. [3]

(ii) Hence, prove by induction that the statement

$$\frac{d^ny}{dx^n} = (n-1)(2e^x) + 2xe^x + \frac{d^{n-1}y}{dx^{n-1}}$$

is true for all positive integers, n , such that $n \geq 2$. [8]

FACTORIAL NOTATION

PROOF HINT

Remember that

$$k! = k \times (k-1) \times (k-2) \times \dots \times 3 \times 2 \times 1$$

and that

$$\begin{aligned}(k+1)! &= (k+1) \times k \times (k-1) \times (k-2) \times \dots \times 3 \times 2 \times 1 \\ &= (k+1) \times k!\end{aligned}$$

$$\therefore (k+1)k! \equiv (k+1)!$$

Example

Question 1

Prove by Induction that

$$\sum_{r=1}^n r(r)! = (n+1)! - 1$$

for all $n \in \mathbb{Z}^+$.

Suggested Solution

When $n = 1$:

$$LHS = 1(1)! = 1 \times 1 = 1$$

$$RHS = (1+1)! - 1 = 2! - 1 = 2 \times 1 - 1 = 2 - 1 = 1 \equiv LHS$$

Since $RHS \equiv LHS = 1 \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\sum_{r=1}^k r(r)! = (k+1)! - 1$$

Then for $n = k + 1$:

$$\sum_{r=1}^{k+1} r(r)! = [(k+1) + 1]! - 1$$

Proof

Note:

$$a) r! = r \times (r-1) \times (r-2) \times \dots \times (r-n); \quad r > n$$

$$b) (r+1)! = (r+1) \times r \times (r-1) \times (r-2) \times \dots \times (r-n); \quad r > n$$

$$\Rightarrow (r+1)! = (r+1) \times r!$$

$$\Rightarrow (r+2)! = (r+2) \times (r+1)! \text{ etc}$$

$$\sum_{r=1}^{k+1} r(r)! = \sum_{r=1}^k r(r)! + (k+1)[(k+1)!]$$

LHS:

$$\begin{aligned} \sum_{r=1}^{k+1} r(r)! &= [(k+1)! - 1] + \{(k+1)(k+1)!\} \\ &= \{(k+1)(k+1)!\} + (k+1)! - 1 \\ &= \{(k+1)[(k+1)!] + (k+1)!\} - 1 \\ &= \{(k+1)! [k+1+1]\} - 1 \\ &= \{(k+1)! \times (k+2)\} - 1 \\ &= (k+2)! - 1 \\ &= [(k+1) + 1]! - 1 \text{ (which is true for } n = k+1) \end{aligned}$$

$\therefore$ The statement is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1 \therefore$ by mathematical induction the statement is true for $n \in \mathbb{Z}^+$.

Question 2

If $y = \frac{1}{1+x}$ then prove by induction that $\frac{d^n y}{dx^n} = \frac{(-1)^n n!}{(1+x)^{n+1}}$ for all positive integral value n .

Suggested Solution

When $n = 1$:

LHS

$$y = \frac{1}{1+x} = (1+x)^{-1}$$

$$\frac{dy}{dx} = -1(1+x)^{-2} = \frac{-1}{(1+x)^2}$$

RHS

$$\frac{(-1)^n n!}{(1+x)^{n+1}} = \frac{(-1)^1 1!}{(1+x)^{1+1}} = \frac{-1}{(1+x)^2}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$

Assuming true for $n = k$

$$\frac{d^k y}{dx^k} = \frac{(-1)^k k!}{(1+x)^{k+1}}$$

Then for $n = k + 1$

$$\frac{d^{k+1} y}{dx^{k+1}} = \frac{(-1)^{k+1} (k+1)!}{(1+x)^{k+1+1}}$$

Proof

$$\frac{d^{k+1} y}{dx^{k+1}} = \frac{(-1)^{k+1} (k+1)!}{(1+x)^{k+2}}$$

LHS:

$$\begin{aligned}
\frac{d^{k+1}y}{dx^{k+1}} &= \frac{d}{dx} \left(\frac{d^k y}{dx^k} \right) = \frac{d}{dx} \left[\frac{(-1)^k k!}{(1+x)^{k+1}} \right] = \frac{d}{dx} [(-1)^k k! (1+x)^{-(k+1)}] \\
&= -(k+1)(-1)^k k! (1+x)^{-(k+1)-1} \\
&= (-1)(-1)^k (k+1)k! (1+x)^{-(k+2)} \\
&= (-1)^{k+1} (k+1)! (1+x)^{-(k+2)} \\
&= \frac{(-1)^{k+1} (k+1)!}{(1+x)^{k+2}} \quad (\text{which is true for } n = k+1)
\end{aligned}$$

$\therefore$ The statement is true for $n = k + 1$

Since the statement is true for $n = 1$, $n = k$, $n = k + 1 \therefore$ by mathematical induction the statement is also true for $n \in \mathbb{Z}^+$.

Question 3

Prove by mathematical induction that

$$\sum_{r=1}^n \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$$

Suggested Solution

When $n = 1$:

LHS

$$\begin{aligned}
\sum_{r=1}^n \frac{r}{(r+1)!} &\equiv \sum_{r=1}^1 \frac{r}{(r+1)!} \\
&= \frac{1}{(1+1)!} = \frac{1}{2!} = \frac{1}{2}
\end{aligned}$$

RHS

$$1 - \frac{1}{(n+1)!} = 1 - \frac{1}{(n+1)!}$$

$$= 1 - \frac{1}{(1+1)!}$$

$$= 1 - \frac{1}{2!}$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\sum_{r=1}^k \frac{r}{(r+1)!} = 1 - \frac{1}{(k+1)!}$$

Then for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{r}{(r+1)!} = 1 - \frac{1}{(k+1+1)!}$$

Proof

$$\sum_{r=1}^{k+1} \frac{r}{(r+1)!} = 1 - \frac{1}{(k+2)!}$$

LHS

$$\begin{aligned} \sum_{r=1}^{k+1} \frac{r}{(r+1)!} &= \sum_{r=1}^k \frac{r}{(r+1)!} + (k+1)th \text{ term} \\ &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+1+1)!} \\ &= 1 + \frac{k+1}{(k+2)!} - \frac{1}{(k+1)!} \\ &= 1 + \frac{(k+1)(k+1)! - (k+2)!}{(k+1)!(k+2)!} \\ &= 1 + \frac{(k+1)(k+1)! - (k+2)(k+1)!}{(k+1)!(k+2)!} \end{aligned}$$

$$\begin{aligned}
&= 1 + \frac{(k+1)! [(k+1) - (k+2)]}{(k+1)! (k+2)!} \\
&= 1 + \frac{[k+1 - k - 2]}{(k+2)!} \\
&= 1 + \frac{-1}{(k+2)!} \\
&= 1 - \frac{1}{(k+2)!} \text{ (which is true for } n = k+1 \text{)}
\end{aligned}$$

∴ The statement holds for $n = k+1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k+1$ then by induction the statement is true for all positive integers n

Question 4

ZIMSEC JUNE 2009 PAPER 2

Prove by Induction that

$$\frac{d^n(x^m)}{dx^n} = \frac{m!}{(m-n)!} x^{m-n}; \quad m \geq n$$

[8]

Suggested Solution

When $n = 1$:

LHS

$$\begin{aligned}
\frac{d^n(x^m)}{dx^n} &\equiv \frac{d(x^m)}{dx} \\
&= mx^{m-1}
\end{aligned}$$

RHS

$$\frac{m!}{(m-n)!} x^{m-n} = \frac{m!}{(m-1)!} x^{m-1}$$

$$= \frac{m \times m - 1 \times m - 2 \times \dots \times 2 \times 1}{(m - 1)!} x^{m-1}$$

$$= \frac{m \times (m - 1)!}{(m - 1)!} x^{m-1}$$

$$= mx^{m-1}$$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$\frac{d^k(x^m)}{dx^k} = \frac{m!}{(m - k)!} x^{m-k}$$

Then for $n = k + 1$:

$$\frac{d^{k+1}(x^m)}{dx^{k+1}} = \frac{m!}{(m - (k + 1))!} x^{m-(k+1)}$$

Proof

$$\frac{d^{k+1}(x^m)}{dx^{k+1}} = \frac{m!}{(m - k - 1)!} x^{m-k-1}$$

LHS

$$\begin{aligned} \frac{d^{k+1}(x^m)}{dx^{k+1}} &= \frac{d}{dx} \left(\frac{d^k(x^m)}{dx^k} \right) \\ &= \frac{d}{dx} \left(\frac{m!}{(m - k)!} x^{m-k} \right) \\ &= \frac{(m - k)m!}{(m - k)!} x^{m-k-1} \\ &= \frac{(m - k)m!}{(m - k) \times (m - k - 1) \times (m - k - 2) \times \dots \times 2 \times 1} x^{m-k-1} \\ &= \frac{(m - k)m!}{(m - k) \times (m - k - 1)!} x^{m-k-1} \\ &= \frac{m!}{(m - k - 1)!} x^{m-k-1} \end{aligned}$$

$$= \frac{m!}{(m - (k + 1))!} x^{m-(k+1)} \text{ (which is true for } n = k + 1 \text{)}$$

∴ The statement holds for $n = k + 1$.

Since the statement holds for $n = 1$; $n = k$ and $n = k + 1$ then by induction the statement is true for all positive integers n where $m \geq n$.

PRACTICE QUESTIONS

Question 1

If $y = \frac{1}{1-x}$ then prove by induction that $\frac{d^n y}{dx^n} = \frac{n!}{(1-x)^{n+1}}$ for all positive integral values n .

Question 2

- (a) Find the first four derivative of $f(x) = -4e^{-2x}$.
- (b) Hence find the formula for $f^n(x)$ i.e. the n^{th} derivative.
- (c) Hence prove your formula by mathematical induction.

Question 3

Prove by mathematical induction that

$$\sum_{r=1}^n (r^2 + 1)r! = n(n+1)!$$

Question 4

It is give that $y = \ln(ax + 1)$, where a is a positive constant.. Prove by mathematical induction that for every positive integer n ,

$$\frac{d^n y}{dx^n} = (-1)^{n-1} \frac{(n-1)! a^n}{(ax + 1)^n}$$

PAST EXAMINATION QUESTION

Question 1

ZIMSEC NOVEMBER 2007 PAPER 2

Given that $y = \frac{1}{ax+1}$, where a is a constant.

- (a) Find expressions for $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$, giving your answers in the form

$$\frac{(-a)^k k!}{(ax + 1)^{k+1}},$$

where $k = 1, 2$ or 3 .

[5]

- (b) Suggest a formula for n^{th} derivative of y in terms of a and n .

[1]

- (c) Prove by Induction the suggested n^{th} derivative of y .

[7]

COMPOSITE FUNCTIONS AND SEQUENCES

Examples

Question 1

(a) Given that $f(x) = \frac{x}{x+1} : x \in \mathbb{R}$, find:

(i) $f^2(x)$ i.e. $f \cdot f(x)$

(ii) $f^3(x)$ i.e. $f \cdot [f \cdot f(x)]$

(b) Suggest a formula for $f^n(x)$

(c) Prove the result in (b) by mathematical induction.

Suggested Solution

(a) $f(x) = \frac{x}{x+1}$

(i) $f \cdot f(x) = f\left(\frac{x}{x+1}\right)$

$$\begin{aligned}\frac{f(x)}{f(x)+1} &= \frac{\frac{x}{x+1}}{\frac{x}{x+1}+1} \\ &= \frac{\frac{x}{x+1}}{\frac{x+1(x+1)}{x+1}} \\ &= \frac{\frac{x}{x+1}}{\frac{2x+1}{x+1}} \\ &= \frac{x}{2x+1}\end{aligned}$$

(ii) $f \cdot [f \cdot f(x)] = f\left(\frac{x}{2x+1}\right)$

$$\begin{aligned}&= \frac{\frac{x}{2x+1}}{\frac{x}{2x+1}+1} \\ &= \frac{\frac{x}{2x+1}}{\frac{x+1(2x+1)}{x+1}}\end{aligned}$$

$$\begin{aligned}
 &= \frac{\frac{x}{2x+1}}{\frac{3x+1}{2x+1}} \\
 &= \frac{x}{3x+1}
 \end{aligned}$$

(b) $f^n(x) = \frac{x}{nx+1}$

(c) When $n = 1$

LHS: $f^n(x) = f(x) = \frac{x}{x+1}$

RHS: $\frac{x}{nx+1} = \frac{x}{(1)x+1} = \frac{x}{x+1}$

Since $LHS = RHS \therefore$ the statement is true for $n = 1$.

Assuming true for $n = k$:

$$f^k(x) = \frac{x}{kx+1}$$

Then for $n = k + 1$:

$$f^{k+1}(x) = \frac{x}{(k+1)x+1}$$

Proof:

$$f^{k+1}(x) = \frac{x}{(k+1)x+1}$$

LHS:

$$\begin{aligned}
 f^{k+1}(x) &= f \cdot f^k(x) \\
 &= f\left(\frac{x}{kx+1}\right) \\
 &= \frac{\frac{x}{kx+1}}{\frac{x}{kx+1} + 1} \\
 &= \frac{\frac{x}{kx+1}}{\frac{x+1(kx+1)}{x+1}}
 \end{aligned}$$

$$= \frac{\frac{x}{kx+1}}{\frac{x+kx+1}{x+1}}$$

$$= \frac{\frac{x}{kx+1}}{\frac{(k+1)x+1}{kx+1}}$$

$$= \frac{x}{(k+1)x+1} \quad (\text{which is true for } n = k+1)$$

∴ The statement is true for $n = k+1$.

Since the statement is true for $n = 1, n = k, n = k+1$ ∴ by mathematical induction the statement holds for $n \in \mathbb{Z}^+$.

Question 2

Given that a sequence $U_1, U_2, U_3, \dots$ is defined by

$$U_1 = 3 ; (n \geq 1) \quad U_{n+1} = 3 - \frac{2}{U_n}$$

Prove by induction that for $n \in \mathbb{Z}^+$

$$U_n = \frac{2^{n+1} - 1}{2^n - 1}$$

Suggested Solution

$$U_n = \frac{2^{n+1} - 1}{2^n - 1} \text{ for all } n \in \mathbb{Z}^+$$

When $n = 1$:

$$U_1 = \frac{2^{1+1} - 1}{2^1 - 1} = \frac{2^2 - 1}{2^1 - 1} = \frac{4 - 1}{2 - 1} = 3 \equiv \text{the given condition } U_1 = 3$$

∴ U_n is true for $n = 1$.

Assuming that U_n is true for $n = k$:

$$U_k = \frac{2^{k+1} - 1}{2^k - 1}$$

Then for $n = k + 1$:

$$U_{k+1} = \frac{2^{(k+1)+1} - 1}{2^{k+1} - 1}$$

Proof

$$U_{k+1} = \frac{2^{(k+1)+1} - 1}{2^{k+1} - 1}$$

Now:

$$U_{n+1} = 3 - \frac{2}{U_n}$$

$$U_{k+1} = 3 - \frac{2}{U_k} \quad (\text{Use the assumption})$$

$$\Rightarrow U_{k+1} = 3 - \frac{2}{\left(\frac{2^{k+1} - 1}{2^k - 1}\right)}$$

$$= 3 - \frac{2(2^k - 1)}{2^{k+1} - 1}$$

$$= \frac{3(2^{k+1} - 1) - 2(2^k - 1)}{2^{k+1} - 1}$$

$$= \frac{3 \times 2^{k+1} - 3 - 2 \times 2^k + 2}{2^{k+1} - 1}$$

$$= \frac{3 \times 2^{k+1} - 3 - 2^{k+1} + 2}{2^{k+1} - 1} \quad (\text{Since } a^b \times a^c = a^{b+c})$$

$$= \frac{(2 \times 2^{k+1}) - 1}{2^{k+1} - 1}$$

$$= \frac{2^{k+2} - 1}{2^{k+1} - 1} \quad (\text{Since } a^b \times a^c = a^{b+c})$$

$$= \frac{2^{(k+1)+1} - 1}{2^{k+1} - 1} \quad (\text{which is true for } n = k + 1)$$

$\therefore U_n$ is true for $n = k + 1$.

Since the statement is true for $n = 1, n = k, n = k + 1 \therefore$ by mathematical induction the statement is also true for $n \in \mathbb{Z}^+$.

PRACTICE QUESTIONS

Question 1

Given that a sequence $U_1, U_2, U_3, \dots$ is defined by

$$U_1 = 5; \quad U_{r+1} = 2U_r + 1, (r \geq 1).$$

Prove by induction that for $n \in \mathbb{Z}^+, U_n = 3 \times 2^n - 1$

Question 2

Given that a sequence $V_1, V_2, V_3, \dots$ is defined by $V_1 = \frac{1}{3}$

$$V_r = V_{r-1} + \frac{1}{r+1}; (r \geq 2)$$

Prove by induction that for $n \in \mathbb{Z}^+$

$$V_n = \frac{n}{n+1}$$

Question 3

For the sequence $U_1, U_2, U_3, \dots$ it is given that

$$U_1 = \frac{1}{4} \text{ and } U_{r+1} = \frac{3U_r - 2}{4}.$$

Prove by mathematical induction that for $n \in \mathbb{Z}^+$

$$U_n = 3\left(\frac{3}{4}\right)^n - 2$$

Question 4

Given that

$$U_{n+1} = 3U_n + 4, \quad U_1 = 1$$

Prove by induction that for $n \in \mathbb{Z}^+$

$$U_n = 3^n - 2$$

Question 5

Given that

$$U_{n+1} = 3U_n + 1, \quad U_1 = 1$$

Prove by induction that for $n \in \mathbb{Z}^+$

$$U_n = \frac{3^n - 1}{2}$$

Question 6

Given that

$$U_{n+1} = 2U_n + 3, \quad U_1 = 7$$

Prove by induction that for $n \geq 1$

$$U_n = 5(2^n) - 3$$

Question 7

- (a) Given that $f(x) = x + 1: x \in \mathbb{R}$, find:
- (i) $f^2(x)$ i.e. $f.f(x)$
 - (ii) $f^3(x)$ i.e. $f.[f.f(x)]$
- (b) Suggest a formula for $f^n(x)$
- (c) Prove the result in (b) by mathematical induction.

TROCKERS

ASANTE SANA

*****THERE IS A LIGHT AT THE END OF EVERY TUNNEL *****

*CONSTRUCTIVE COMMENTS ON THE FORM
OF THE PRESENTATION, INCLUDING ANY
OMISSIONS OR ERRORS, ARE WELCOME.*

*****ENJOY*****

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